COGNITIVE WARFARE

The Battlespace of the Mind

EXCLUSIVE
FOREWORD

Cognitive WARFARE

The Battle for Minds

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IN TODAY'S UNCERTAIN and fast-evolving strategic environment, we find ourselves under constant siege as cognitive war rages. It is the stark reality of contemporary conflict, where traditional boundaries between peace, crisis and war have evaporated and where the human mind is a primary target. Every hour of every day, our adversaries are expending time and effort in order to fragment our societies, sow doubt and undermine NATO unity and cohesion.

"**War is here**, lurking in the **grey zone**, and victory requires nothing less than our complete **commitment to defending the minds and values** that define who we are as free nations."

Recent studies estimate that Russia spends up to 2 billion U.S. dollars annually on cognitive warfare operations,[1] funding destructive social and psychological manipulation campaigns, including sophisticated disinformation, conspiracy theories, and large-scale bot networks.

At Allied Command Transformation (ACT), we recognize that cognitive warfare represents far more than an evolution in information operations. It constitutes a paradigm shift that challenges our most fundamental assumptions about deterrence and defence.

In his OODA loop (observe, orient, decide, act), John Boyd pointed to how uncertainty and confusion in an adversary's decision-making cycle can produce paralysis and bring about collapse.

Today's cognitive warriors have weaponized this insight,[2] systematically targeting the orientation phase to degrade our collective judgment, distort our perceptions, and ultimately compromise our ability to act decisively in defence of our shared values.

The brain has become a new war zone, both as a target and a weapon. Our adversaries understand that a cognitive knockout can forestall wins in the land, maritime, air, cyberspace or space domains. They therefore seek to contest our decision space, manipulate our populations, and ultimately deprive us of our freedom of choice through sustained campaigns that operate in the grey zone, below the traditional threshold of armed conflict.

This reality demands a mental shift in how we conceive of security challenges. We often speak of cognitive warfare as a future concern, warning that "we will lose the fight when it comes." But the fight is here. The question we must ask ourselves — as military leaders, as Alliance members, as defenders of democratic values — is whether we are ready to participate in a war that is already underway. Can we defend our cognitive ground? Can we shape the perceived battlespace of our adversaries?

ACT'S JOINT WARFARE CENTRE has long served as NATO's training focal point for both operational- and strategic-level exercises, supporting the Alliance through the injection of new concepts and doctrines in exercises. Today, operationalizing NATO's Cognitive Warfare Concept represents one of our most critical contributions to Alliance defence. Without cognitive superiority, our conventional deterrence and defence capabilities risk becoming irrelevant. We could lose battles before they have even begun.

The articles in this edition of The Three Swords examine these challenges from multiple perspectives, offering insights from practitioners, researchers, and strategic thinkers who are grappling with the practical implications of cognitive warfare. From tactical applications in military operations to strategic communication challenges, from ethical considerations to technological solutions, these contributions reflect the breadth and complexity of weaponized cognition.

Meeting these challenges requires mental agility, critical thinking, and human expertise applied with unprecedented coordination across the Alliance. It calls for innovation in how we train our forces, educate our fellow citizens, and defend our information environment. Most importantly, it requires trust, unity of purpose and action across the whole Alliance.

THE SCALE AND SOPHISTICATION of cognitive threats targeting NATO cannot be overstated. Every social media platform, every news cycle, every public debate has become a potential battleground where truth competes with fabrication, where democratic discourse faces systematic manipulation, and where the very foundations of informed citizenship are under assault. But we are not passive victims in this contest. ACT is constantly on the lookout for future challenges and develops the capabilities necessary to address them. Together, we can build the cognitive defences our Alliance requires while maintaining our commitment to open societies and democratic values.

This issue concerns not only our military forces but society as a whole. It is a challenge that requires strong commitment and determination from political leadership, civil institutions, the private sector, and every individual. By acting together decisively and purposefully, will we ensure that cognitive superiority remains firmly within NATO's grasp. The future of our Alliance, of our collective strategic interests and core values, may well depend on how effectively we meet this challenge.

As you engage with the ideas presented in this review, remember that cognitive warfare is not merely an academic exercise or a distant strategic concern. It is the defining security challenge of our time, one that demands both intellectual rigor and practical action from every member of the Alliance community.

The time for half-measures has passed. War is here, lurking in the grey zone, and victory requires nothing less than our complete commitment to defending the minds and values that define who we are as free nations. ✦

ENDNOTES

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2 The Dialectic of Deception: John Boyd and the Cognitive Battlefield, War on the Rocks, 4 September 2025 https://warontherocks.com/2025/09/the-dialectic-of-deception-john-boyd-and-the-cognitive-battlefield/

The Battlespace of the Mind:
Command, Control, and the Cognitive Frontier

"New – not only physical – spheres of confrontation with new types of weapons are appearing, for which the priority is not physical, but functional defeat of the enemy... This comprehensive approach to the impact on the enemy in modern war is becoming an objective necessity."[1]

Colonel General Vladimir Borisovich Zarudnitsky,
Chief of the Russian Military Academy of General Staff

"Psychological warfare has become an indispensable operational form in a modern campaign... [it] has a powerful 'soft kill' ability in war [that] can achieve effects other operational activities cannot."[2]

The Science of Campaigns,
Beijing National Defence University

IN TODAY'S CONFLICTS — and those of the future — the fight for strategic advantage is waged not only on the battlefield, but within the human mind. Both Russia and China recognize that influencing beliefs, behaviours, and command and control structures can be as decisive as control over physical terrain or infrastructure. The emergence of new technologies is amplifying existing risks. Ten years ago, Russia's "little green men" were at the cutting edge of deception operations. Today, disruptive technologies such as artificial intelligence (AI), machine learning, deepfakes, and biotechnology and human enhancement (BHET) are supercharging the ability to influence, deceive, and cognitively overload military and civilian leadership — enhancing the scale, precision and effectiveness of such operations.

This article has been prepared by Headquarters Supreme Allied Commander Transformation, Concept Development Branch. All views expressed are those of the branch and do not necessarily represent those of the Alliance or member nations.

THIS ARTICLE EXPLORES the doctrinal foundations underpinning Russia's and China's approaches to the cognitive dimension, with a particular focus on how both states are developing and adapting their capabilities to disrupt adversary military command and control (C2) through cognitive operations. By examining the evolution of these strategies, this article aims to illuminate what Russia and China regard as a critical enabler of strategic success: the systematic targeting and degradation of decision-making architectures through the integrated use of conventional military instruments of power and non-military instruments of power.

Maskirovka, Reflexive Control, and New Generation War: The Russian Battlespace of Perception

Maskirovka, a cornerstone of Russian military tradition, refers to strategic deception designed to mislead an opponent about Moscow's true intentions and compel poor decisions that ultimately benefit Russia.[3] It laid the groundwork for Russian psychological operations, which emerged as a critical tool during the Russian Civil War (1917–1922). During this period, Bolshevik commissars within the Red Army's Main Military-Political Directorate managed internal political education, morale boosting and psychological warfare targeting enemy troops and civilians. Lenin emphasized the internal "disintegration" of enemy forces over "external warfare" (or, in other words, conventional warfare), with early examples including camouflage and feints during the Russo-Japanese War.[4] Over time, this approach evolved, adapting to the technological and geopolitical realities of each era — from WWII leaflet drops to modern cyber "hack-and-leak" operations. *Maskirovka*'s enduring relevance lies in its fusion of deception, psychological manipulation and intelligence gathering.

"Russia doctrinally treats the cognitive dimension as a full-spectrum battlespace."

First theorized by Soviet mathematical psychologist Vladimir Lefebvre in the 1960s, reflexive control (RC) builds on Soviet-era game theory and Marxist-Leninist epistemology, treating cognition as deterministic and manipulable.[5] RC is the art of subtly influencing an adversary's decision-making by feeding them carefully crafted information so that they "voluntarily" make choices that favour the controller. Unlike basic deception, RC requires modelling the adversary's cognition process, anticipating not just behaviour, but reasoning. Biographical data, habits, and psychological deficiencies become exploitable variables.

As Russian theorists stress, success comes from understanding how the target thinks, then reinforcing that logic to engineer strategic self-defeat.[6] RC is therefore conceived of as longer-term operations specifically aimed at shaping an adversary's perceptions and choices (without their awareness) to achieve asymmetric military effect before kinetic action is required.

Analytically, RC is divided into:

- **Constructive or "friendly embrace"[7] RC,** which induces complacency, hesitation and paralysis in the target's decision-making and creativity — e.g. Putin's ceasefire with Poroshenko prior to the Russian invasion in 2014.[8]
- **Destructive RC,** which incapacitates the target's decision-making through information overload and the weaponization of psychological deficiencies and cognitive weaknesses through a carefully tailored deception operation — e.g. Russia's hack of ViaSat's satellite network to disrupt Ukrainian command and control during its full-scale invasion on 24 February 2022.

RC thus treats information as both technical data and cognitive and emotional content.

Modern command and control architecture spans three decision-making modes: human-only, fully automated machine-only, and, most commonly in current military practice, hybrid human-machine collaboration. Each presents distinct vulnerabilities to adversary manipulation. Adversaries can target either the cognitive aperture of the commander or the technical substrata that feed and assist that commander. The injection of false, irrelevant, or mistimed inputs across either vector can degrade decision quality and tempo.

Crucially, according to retired Major General M.D. Ionov, an early proponent of RC in Russian military thought, "information" in this context encompasses not merely raw data but also the emotional cues and control elements that frame perception and authority.[9] For instance, a show of military force may be intended less to display troop strength and equipment, and more to intimidate or provoke strong emotional reactions.

With these vulnerabilities firmly in mind, Russia doctrinally treats the cognitive dimension as a full-spectrum battlespace that can precede or even substitute for military operations. In particular, the contemporary Russian concept of new generation warfare builds upon *maskirovka* and reflexive control by conceiving of a complex application of military, economic, social, and political methods to influence adversaries.[10]

Beyond shaping perceptions, these actions seek to reconfigure an adversary's thought processes and behaviours at individual and societal levels, ultimately undermining their will to resist.

Russia seeks cognitive dominance by targeting the adversary's decision-making cycle, treating it as the primary centre of gravity. RC campaigns follow a five-phase model: (1) shape the perceptual environment; (2) constrain options; (3) manipulate perceptions; (4) exploit induced errors; and (5) consolidate gains.[11] This effort is executed through a diverse ecosystem of actors: dedicated information operations units, state media, intelligence agencies, and cyber forces (notably, the Yevgeny Prigozhin-founded Internet Research Agency).

In parallel, Russia harnesses non-state entities ranging from private military companies such as Wagner, Redut, and Patriot, to criminal networks such as Salem, Bashkaki, and the ultranationalist Night Wolves motorcycle club (also known as "Putin's Angels"), as well as the notorious Solntsevo network. Moscow also sponsors proxy groups, including pro-Kremlin separatists such as the Russian Imperial Movement (RIM) and religious militias such as the Russian Orthodox Army (ROA) to advance its strategic aims.[12]

All act in concert to confound attribution and blur the line between deterrence, deception and coercion. In practice, Moscow implements cognitive warfare through coordinated psychological operations, electronic warfare (EW), cyber attacks, and misinformation campaigns. In doing so, Russia exploits all three C2 modes:

- **Human-only,** through psychological operations (propaganda, reflexive control, indoctrination) that target commanders directly — manipulating perception, inducing misguided judgment, and suppressing dissent;

- **Machine-only,** through disinformation, spoofing, and cognitive cyber attacks that corrupt data inputs and degrade algorithmic decision-making across automated systems;

- **Hybrid systems,** through EW,[13] such as Divnomorye and Murmansk-BN complexes, and cyber operations that inject corrupted, mistimed, or false data — skewing human-machine collaboration, distorting situational awareness, and paralyzing decision cycles.

Operationally, the Kremlin seeks to interdict decision-making through phased campaigns that open with covert shaping and escalate only if resistance solidifies, utilizing tactics long forgotten by the West. Preempting that, peacetime disinformation and intimidation operations are designed to fracture domestic consensus and delay critical decision-making.

During territorial grabs in Crimea and the Donbas, for example, Moscow combined unmarked troops with denial, cyber disruption and PsyOps to paralyze Kyiv's and NATO's response cycles, presenting a fait accompli before political authorities were able to respond.

These measures sit inside a broader escalation ladder[14] that combines psychological shocks with the implied readiness to employ precision fires and, ultimately, nuclear coercion should other cognitive blows fail. The net effect is to degrade cohesion, slow target decision-making and undermine the will to resist — ideally long before Russian troops cross the border, but with operations continuing in subsequent phases as well.

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Maskirovka and Russia's Seizure of Crimea

In 2014, masked men in unmarked uniforms appeared on the Crimean Peninsula. To locals, they looked innocuous. Crimean self-defense forces? Maybe police? Certainly not the vanguard of a foreign invasion. In Kyiv, Ukrainian leaders debated taking action but hesitated; they saw the Russian troops but could not pin down their intent.

Memories of Georgia in 2008 loomed large — any misstep might trigger all-out war. So, they exercised restraint. Meanwhile, in Western capitals, officials urged calm, mistaking Russia's deception for a search for compromise, not conquest. By the time the lie became too big to ignore, Crimea was in Russian hands.

This was *maskirovka*, Russia's century-old art of military deception, executed with chilling precision. Moscow demonstrated that it did not need overwhelming force to achieve its aims, but only the ability to sow confusion, instill doubt, and create delays; paralyzing decision-making at tactical, operational, and strategic levels.

Winning Without Fighting: China's Cognitive Warfare Doctrine and Civil-Military Integration

Like Russia, China sees cognitive warfare as a central pillar of modern military operations — employed across peacetime, crisis, and conflict to disrupt adversary decision-making. At the 20th Party Congress in 2022, Chinese leaders explicitly called for the "study and mastery of the characteristics and laws of information-based and intelligent warfare," underscoring the centrality of cognitive and informational dominance in contemporary Chinese strategy.[15]

This approach reflects both classical and modern currents in Chinese strategic thinking. Ancient theorists like Sun Tzu emphasized subduing the enemy through deception, disorientation, and manipulation.[16] Philosophical traditions such as Confucianism and Mohism advanced the notion that moral authority and persuasion constitute more enduring and legitimate forms of power than brute force. In the modern era, Mao Zedong developed a distinct framework rooted in Marxist-Leninist revolutionary doctrine, placing heavy emphasis on propaganda, ideological control, and psychological mobilization to consolidate internal unity and erode enemy morale. During the Korean War (1950–1953), these principles were operationalized through propaganda campaigns aimed at eliminating "pro-American ideology" and strengthening resilience among both troops and civilians.[17]

In the 21st century, these ideas were first crystallized into the People's Liberation Army's (PLA) "Three Warfares" doctrine, comprising both internally and externally directed efforts to conduct[18]:

- **Psychological Warfare:** targets military commanders and troops;
- **Public Opinion Warfare:** shapes perceptions of domestic and international audiences;
- **Legal Warfare:** leverages international law to legitimize Chinese actions and delegitimize adversaries.

China's "Three Warfares" doctrine, in turn, is deeply embedded within the PLA's broader strategic trajectory as it seeks to become a "world-class military" by 2049, the centenary of the founding of the People's Republic of China. The PLA's modernization trajectory is envisioned through three progressive phases: mechanization, informatization, and intelligentization.[19]

Mechanization, dominant through to the first decade of the 21st century, focused on building a modern, industrial-era military with advanced hardware such as tanks, aircraft and naval vessels.

Informatization, the current phase of China's military modernization process spanning from around 2010 to the late 2020s, emphasizes integrated command, control, communications, computers, intelligence, surveillance, and reconnaissance (C4ISR) to enable joint operations and data-driven warfare across domains.

In the emerging phase of **intelligentization**, to span the late 2020s through to 2049, the PLA is integrating emerging technologies such as artificial intelligence, big data, autonomous systems, human-machine teaming, and research into brain science to reshape the battlefield, aiming to outpace adversaries not just through force but by achieving faster, more adaptive decision-making and cognitive dominance.

Chinese strategists adopt a broad, systemic approach to cognitive warfare, deliberately erasing the boundaries between military and civilian domains in a whole-of-society effort. The authoritative Science of Military Strategy, published by the PLA's Academy of Military Sciences, characterizes psychological warfare as a "full-dimensional strategic action" that spans "politics, economy, military affairs, diplomacy, culture and religion."[20] This strategy serves dual purposes: to bolster domestic psychological resilience against foreign influence — driven in part by the leadership's enduring fear of a Soviet-style collapse linked to ideological erosion — and to degrade an adversary's capacity (at an individual or collective level) to resist cognitive manipulation. This strategy rests on several interlocking pillars:

1. Civil-Military Integration: China's national strategy mandates the integration of civilian and military research, infrastructure, and expertise. Tech companies, universities, media outlets, and cultural institutions are expected to contribute to information warfare campaigns, whether by developing AI-enabled cognitive tools, amplifying Party narratives, or conducting influence operations abroad.

2. Information Control and Narrative Dominance: The CCP maintains tight control over domestic information flows through censorship, surveillance, and propaganda. At the same time, it increasingly engages in global public opinion warfare — via state media (e.g. CGTN, Xinhua), social media bots, and United Front operations — to frame international discourse in terms favourable to Beijing.

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Three Warfares and South China Sea Disputes

In January 2024, a phone call took place between Vice Admiral Alberto Carlos, then head of the Philippines' Western Command, and Senior Colonel Li Jianzhong, China's defence attaché. No announcement followed immediately, but months later, the Chinese Embassy claimed that this call resulted in a new agreement over the contested Second Thomas Shoal — a so-called "new model" to ease tensions. Beijing released what it said was a transcript and played a portion of the alleged recording for journalists, framing the exchange as proof that Manila had quietly conceded to Chinese terms. Carlos confirmed the call but denied any agreement had been made or that he had consented to being recorded. Foreign Secretary Enrique Manalo and Defence Secretary Gilberto Teodoro Jr. likewise rejected Beijing's claims, insisting that there had never been a Cabinet-level consensus on any Chinese proposal for the shoal. Yet the damage was done. By publicizing a murky, unconfirmed exchange and presenting it as a diplomatic concession, China weaponized ambiguity to shape perception. It didn't need an actual deal — only the suggestion of one.

3. Psychological Conditioning of the Population: The CCP actively shapes public sentiment and emotional resilience through patriotic education, ideological training, and elaborate propaganda. This seeks to ensure not only national unity in times of external crisis, but also primes the population to psychologically withstand external narratives and support the strategic objectives of the state.

4. Legal Warfare and Norm Manipulation: Chinese scholars and officials reinterpret international law to suit strategic goals — asserting, for instance, that certain legal "gray zones" justify Chinese activities in the South China Sea or Taiwan Strait. These arguments are pushed across diplomatic, media, and academic platforms to wear down the legitimacy of adversary actions and create confusion in international forums.

Operationally, China's cognitive warfare may seek to target C2 at all levels:

- **Senior Leaders:** to induce doubt, paralysis, and strategic misjudgment;
- **Military Personnel:** to weaken discipline, cohesion, and operational tempo;
- **Civilian Populations:** to sow distrust in government and generate internal pressure on decision-makers.

By corrupting and degrading information flows, overwhelming decision-support systems with manipulated or contradictory data, and fostering psychological fatigue among leadership and operators alike, China aims to paralyse adversary decision-making cycles at their core and, if possible, alter decision-making in its favour. These operations aim not just to delay or disrupt, but to induce a persistent state of confusion, doubt, and hesitation, undermining the ability to act decisively at critical moments.

Crucially, they are not confined to the military sphere. By blurring and ultimately erasing the boundary between civilian and military domains, China targets political leaders, military commanders, media systems, and societal institutions as parts of an integrated battlespace. The objective is strategic paralysis: to erode the psychological, informational and institutional foundations of resistance long before conflict begins, degrading a target's capacity to mobilize, coordinate, or respond effectively to aggression or coercion.[21]

China continues to build institutional capacity and organizational coherence around its cognitive warfare strategy.

The PLA's Base 311, originally established to conduct psychological operations against Taiwan, has evolved significantly in recent years. Once reliant on traditional tools such as radio broadcasts and leaflet drops, the base has transitioned to more sophisticated modes of influence, including coordinated digital platform activity, algorithmic amplification of pro-CCP content, and the use of AI-generated disinformation. Its operations now target not only Taiwanese military and political actors but also civil society, journalists and the diaspora, aiming to fracture internal cohesion and erode trust in democratic institutions.[22]

This operational shift is mirrored at the structural level. Recent reforms have radically reconfigured China's cognitive and information warfare apparatus. In 2024, Beijing dissolved the Strategic Support Force — previously the umbrella organization for space, cyber, and electronic warfare capabilities — and replaced it with more specialized and streamlined entities: the Information Support Force, Cyber Support Force, and Aerospace Force. These new forces report directly to the Central Military Commission, ensuring tighter political control and improved alignment with top-level strategic priorities. Their integration into theatre commands enhances their responsiveness and ensures that cognitive operations are embedded in broader joint and multi-domain campaigns.

Emerging Technology and the Future of Cognitive Warfare

While cognitive warfare has long been a feature of both Russian and Chinese military thought, emerging technologies are now supercharging these strategies. Though their approaches differ in scope and sophistication, both states view technological innovation as a pathway to gaining asymmetric advantage.

Moscow is investing in emerging technologies to offset its demographic challenges and leveraging modern capabilities that can more effectively target C2. Facing a shrinking pool of military-aged personnel, Russian defence thinkers advocate for the development of an "army of robots," pivoting from workforce-intensive forces to autonomous and semi-autonomous systems.[23] This vision is reflected in a diverse range of military technology developments, albeit with varying levels of maturity and credibility. Key initiatives include:

- **Robotic and uncrewed platforms:** Systems such as the Marker UGV, Okhotnik heavy combat drone, and humanoid robots (Teledroid, Tester, and Fedor) showcase Russia's emphasis on autonomous capabilities for ISR, targeting, and battlefield disruption.
- **Enhanced soldier systems:** Programmes such as Sotnik, Ratnik, and Legionnaire aim to augment soldier effectiveness through sensor integration, robotic subsystems, AI, and digital battlefield connectivity, with en masse fielding expected by 2035.[24]
- **Neurotechnology and human enhancement:** Russia is exploring brain-computer interfaces (BCIs), neural helmets, and neuropsychological diagnostics. While these remain in early stages, they reflect a strategic intent to blur the boundary between human cognition and machine control.
- **Immersive training and simulation:** Investments in augmented/virtual/mixed reality-based training environments aim to increase preparedness and reduce operational risk.

Of course, it may be the case that some initiatives serve more as demonstrations (or even *maskirovka*) than as operational capabilities. Moreover, systemic barriers — including limited access to foreign technology and sanctions — will constrain Russia's ability to scale or operationalize its most advanced systems. In the near term, Russia is expected to prioritize cost-effective EW systems, implement moderate upgrades to legacy platforms, and deploy modestly advanced C2-disruptive technologies.

Through its doctrine of intelligentized warfare, Beijing views emerging technologies as central to achieving military dominance, including in the cognitive dimension. To this end, China has made technological leadership and indigenous innovation the centrepiece of its national rejuvenation agenda, as outlined

in successive five-year plans. Motivated by a history of technological vulnerability and the desire to avoid the "dangers of falling behind," Chinese leaders have prioritized self-reliance and rapid advances in disruptive technologies such as AI, quantum computing and 5G.

Xi Jinping's "innovation-driven development" strategy is designed not only to close gaps with leading powers but also to seize first-mover advantages where possible; for instance, investing heavily to secure leadership in critical sectors such as 5G, commercial drones, offensive hypersonic weapons, and lithium battery production. Regarding the cognitive dimension specifically, China is undertaking the following initiatives:

- **AI and machine learning:** AI is central to China's cognitive warfare strategy, enabling rapid data analysis, automated sentiment analysis and the creation of tailored influence campaigns. Strategic megaprojects, such as the New Generation AI Development Plan, are intended to make China the global leader in AI by 2030.[25]

- **Brain science and neurocognitive warfare:** Chinese strategists view the brain as the "main battlefield" of future warfare.[26] The PLA is investing in "brain science"[27] to understand and potentially manipulate human cognition, emotions, motives and judgments. Theoretical ambitions include directly interfering with or subconsciously controlling enemy brains to induce confusion, hallucinations or mental fatigue. For instance, the "China Brain Project" is a national initiative aiming to decode brain function and develop ways to enhance or disrupt cognitive processes.

- **Biotechnology and human enhancement (BHET):** China is developing gene editing, synthetic biology and brain-computer interfaces, potentially to produce cognitively enhanced soldiers and degrade adversary cognition. Although many of these technologies remain in experimental phases, over 150 military-related BHET projects are underway in China.[28]

China's dominance in emerging and disruptive technologies (leading in 37 of 44 key fields)[29] positions it to target military C2 architectures through asymmetric, cross-domain operations. While not all capabilities are fully operational, China's rapid progress, state-backed research and development pipeline, and dual-use technology strategy present a formidable challenge. Underpinned by their 2022 declaration of a "no-limits" friendship,"[30] Moscow and Beijing have committed to expanding collaboration across several high-tech sectors, including 5G telecommunications, AI, biotechnology, and the digital economy.[31] Growing institutional ties and policy support for joint projects and forums suggest a strategic push towards coordinated development in technologies that could significantly enhance each state's cognitive warfare capabilities.

Cognitive warfare is not simply a risk to manage; it is a strategic dimension to master.

Early signs of this cooperation include shared academic research on brain function and AI integration. In 2019, a delegation from the Russian Academy of Sciences visited laboratories in Shanghai and praised China's advances in neuromorphic intelligence and brain research.[32] In the media space, the creation of the world's first Russian-speaking AI news anchor — developed through a partnership between state-owned media organs China's Xinhua and Russia's TASS — highlights efforts to fuse AI and propaganda tools.[33]

Russia's deep experience in psychological operations could be further empowered by China's strengths in big data, sentiment analysis and AI. Likely areas of future collaboration include social media manipulation, dual-use neuroscience, emotion recognition, and predictive behavioural modeling. Joint efforts may also extend to synthetic media, such as deepfakes, and immersive digital ecosystems like the metaverse, designed to shape narratives, guide perception and project influence at scale.

Conclusion

Cognitive warfare poses a persistent and escalating threat to military and political command and control systems, as adversaries and challengers exploit the blurred boundaries between civilian and military spheres across borderless dimensions. But these tactics should not be seen merely as threats to neutralize — they are proof of concept: strategic advantage can be won through influence rather than force alone.

This demands a fundamental shift in mindset, from countering disinformation to operationalizing influence; from protecting decision-making to mastering its manipulation;

"Cognitive warfare demands a fundamental shift in mindset, from countering disinformation to operationalizing influence; from protecting decision-making to mastering its manipulation; from building resilience to seizing initiative."

from building resilience to seizing initiative. We should study and adapt the methods of cognitive warfare practitioners. Russia and China have long treated the mind as a battlespace, seeking to wield perception, psychological disruption, and narrative control. Through sustained experimentation, they have embedded cognitive operations into the core of their strategic arsenals, blending military, informational, economic, and political tools to fracture cohesion, distort reality, and disable decision-making — capabilities now supercharged by emerging technologies.

The reality of contemporary conflict is that it often begins not with kinetic strikes but with narratives, symbols and seemingly innocuous shifts in public sentiment. Left uncontested, these shape the very conditions of war before a single round is fired. The lesson from adversaries and challengers is clear: it is not enough to defend our own cognitive space. We must seek cognitive advantage and strive towards cognitive superiority.

Cognitive warfare is not simply a risk to manage; it is a strategic dimension to master. Those who understand and shape perception will hold the initiative across the entire continuum of peace, crisis, and conflict.

NATO must act decisively to avoid being outmanoeuvered. ✧

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THE BRAIN IS BOTH THE TARGET AND THE WEAPON

THE STRATEGIC NECESSITY OF NATO'S COGNITIVE WARFARE CONCEPT

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The views and opinions expressed in this article are those of the author and do not necessarily represent the official position or policy of member governments or of NATO.

"Cognitive warfare is not merely an academic exercise or a distant strategic concern. It is the defining security challenge of our time, one that demands both intellectual rigor and practical action from every member of the Alliance community."

Admiral Pierre Vandier
Supreme Allied Commander Transformation

The War Before the War

On August 21, 2013, the world awoke to haunting images from Ghouta, a suburb of Damascus. Civilians — many of them children — were gasping for air, convulsing, and dying in the streets. The evidence pointed to a massive chemical weapons attack by the Assad regime, crossing the "red line" set by U.S. President Barack Obama only a year before.

Almost immediately, the information space was filled with competing narratives. Russian media outlets, diplomats, and proxy influencers, supported by botnets and troll farms, cast doubt on what had happened. Was it a false flag? Could the footage have been staged? In days, what appeared to be a clear-cut atrocity dissolved into ambiguity. These narratives, carried across RT, Sputnik, fringe Western outlets, and social media platforms, tapped into an old wound rooted in public mistrust after the false weapons of mass destruction claims preceding the Iraq War.

This was more than propaganda. It was a deliberate campaign to muddy intelligence, inject doubt into Allied debates, and stall action. Washington hesitated, then backed away from strikes, accepting instead a Russian-brokered deal.[1] Russia's approach in Syria remains one of the clearest examples of how cognitive warfare can target military command and control (C2) at both tactical and strategic levels. By 2015, Russia had paired these narrative operations with electronic and cyber tools — spoofing communications, jamming UAV feeds, and disrupting intelligence, surveillance, and reconnaissance (ISR) — to erode commanders' ability to act decisively. These were not incidental effects but the intended outcomes of an integrated cognitive campaign: victory achieved not through combat, but through paralysed decision-making before the fight began.

The lessons of Ghouta echo today. At the 2025 NATO Communicators Conference in September, participants underscored that adversaries are no longer simply contesting the information space; they are deliberately targeting the Alliance's ability to think, decide, and act. The message was clear: NATO must start taking the cognitive threat seriously, treating it as a contested domain in its own right.

However, this requires more than words. It requires investment in people, training, and capabilities on a scale reflecting the priority adversaries place on cognitive effects. Until NATO develops the ability not just to defend but to contest in this space, it risks repeating the same paralysis that followed Ghouta. That danger captures the essence of cognitive warfare — but to understand it, we must first define it.

What Is Cognitive Warfare?

Despite growing thought leadership in this space, a comprehensive understanding of cognitive warfare remains elusive. It is often mistakenly equated with hybrid warfare; however, while hybrid warfare involves the coordinated use of multiple instruments of power, with information as one of many tools, cognitive warfare differs in its core objective: to target and influence both human and machine cognition. It can be pursued through any domain, by any means, and at any stage in the continuum of competition, with the aim of obscuring truth, inducing decision paralysis, and shaping perceptions and behaviour.

Cognitive warfare is also not a replacement for strategic communications. In this space, information operations (InfoOps), psychological operations (PsyOps), and military public affairs (MilPA) are just a few of the capabilities and functions employed in the daily contest for cognitive advantage. As such, cognitive warfare is not the means by which we fight; it is the fight itself.

This fight is not simply about the proliferation of disinformation or propaganda, nor is it about time-honoured deception and trickery. Cognitive warfare is the deliberate targeting of human and machine cognition to influence how people think, what they feel, and, ultimately, how they act.

In this battlespace, the brain is both the target and the weapon — the terrain and the conduit through which strategic outcomes are won or lost. Emotions, narratives and identities can be engineered en masse and disseminated to others to alter the course of public opinion, destabilize societies, and influence critical decisions, frequently below the threshold of armed conflict. Though seemingly abstract, many of these operations should be viewed as cognitive attacks: orchestrated information activities designed with hostile intent to manipulate perceptions, beliefs, objectives, decisions and behaviours.

However, understanding cognitive warfare solely through a military lens is insufficient. To fully grasp how it functions, the starting point is recognizing its impact on us as individuals, something that does not come easily. After all, cognitive warfare is, at its core, about humans — and humans are notoriously complicated and volatile, making it problematic to predict behaviour, even our own.

How Cognitive Warfare Impacts Us Individually

The challenge for many is that the ambiguities inherent in the cognitive dimension make it difficult for us to see and understand how cognitive warfare operates, particularly at a daily micro level.

However, you only need to walk into a restaurant or café to see the cognitive battlefield. Entire families sit silently at tables — not talking, just scrolling social media feeds comprised of other people's lives, thoughts and opinions. Many of us fall prey to this technological trap because scrolling and posting are often easier than engaging: we're burned out and tired; real-life relationships are hard; and technology is at our fingertips, providing feel-good dopamine surges. This perfect storm of emotional exhaustion, digital ease, and pleasurable brain chemicals makes it easy to believe that our reliance on constant stimulation is harmless. But it is not.

Today's information environment (IE) is intentionally designed to be addictive, to keep us reading headlines, sharing memes, and watching videos. Billion-dollar industries are built upon platforms and algorithms orchestrated to monetize our propensity to check out of our private lives by checking in to social media. However, our increasing digital immersion and fixation on outrage, opinions and chaos is not benign. It is not merely a cultural shift altering how we socialize; it is a strategic vulnerability that primes us for adversarial influence.

In previous generations, people sat with their thoughts. They spent time thinking without distraction in quiet moments, giving their brains the space to have big ideas. But we no longer sit with our thoughts. We sit on our devices, letting the thoughts of others influence us, all while believing we are thinking for ourselves. We outsource our attention and fill our idle moments with a constant stream of memes, headlines, and outrage. The problem is not just distraction; it is infiltration. Because much of the information we consume is not neutral, and what we think are our own thoughts are often just the opposite.

We are being targeted, not randomly but deliberately, and we rarely recognize that the information we consume is a threat. We do not see when it alters our perceptions because we think we have changed our minds based on our own logic. However, often, those changes in belief are unconscious and the result of emotions manipulated by malign actors who benefit from our anger and frustration.

The challenge is that addressing cognitive warfare requires each of us to examine our own behaviour, whether we are civilians, military members, or government officials. It is very easy for us to say:

> *"So, what? I'm on my phone."*
> *"My watch tracks my steps."*
> *"I decompress by scrolling X. My kids enjoy YouTube. TikTok makes my wife laugh. What's the big deal?"*

But, if we step back, what does this dynamic, which stretches across nations and cultures, say about the direction of our societies? And what risks does this persistent access to our data, from our shopping habits and proclivities to our heart rates, pose to us as individuals, leaders, and nations? More importantly, it raises the question: to whom are we giving our time and attention? Do we even know? And what opportunities are we providing adversaries — through influencers, memes, bots, and deepfakes — to shape what we think and how we behave? For most of us, the answer is: we don't know.

"To whom are we giving our time and attention? What opportunities are we providing adversaries — through influencers, memes, bots, and deepfakes — to shape what we think and how we behave? For most of us, the answer is: we don't know."

Why Cognitive Warfare Works

A malign actor who understands the emotional power of symbols such as flags or religious texts will use them to spread their ideas, evoke emotions, and influence people to act in their favour; even better if those symbols can be used to signal a call to action for specific groups. This malign actor can then use information — including knowledge about individuals, how communities are structured, and the meaning of historical events in target populations — to prompt people to react reflexively because they know that we humans are not the rational actors we believe ourselves to be. And that is our primary vulnerability.

While there is debate about the statistics, only 2–15% of human thinking is driven by logical reasoning, with the remaining 85–90% influenced by emotions, instincts, and unconscious processes.[2,3] It is that 85–90% share that makes us reactively click the headline, share the meme, rage about the story, and retreat into digital echo chambers, often without clear awareness.

However, as a society, we continue to overlook the signs that we are being manipulated because this is not about a tangible battle for borders or territory; this is about an intangible battle for our minds. And whilst there is variation in the extent, no one is immune, regardless of rank, education level, age, or IQ.

If I can evoke an emotional reaction in you based on carefully tailored information and imagery, I can also elicit a similar response in others like you. If I can then create a message that rapidly spreads throughout already disenfranchised populations via social media, I can trigger uprisings, movements, riots, and discord. I can make you distrust your systems, hate your leaders, dismiss your family members, and fear your neighbours. And I do not have to use the truth. I can create fake AI-generated videos, fictitious speeches, and fabricated events. I only need to know which symbols will resonate with specific populations, which narratives will tap into existing fears or shame, and which audiences are already vulnerable to my influence. Mere fact-checking will not usurp my ability to guide your perception of reality.

From Targeting Populations to Timing Perception

Skeptics often argue that cognitive warfare is not new, and it isn't. Militaries and governments have used weaponized information and targeted propaganda for centuries. However, what *is* new is the pace of change and the ability of malign actors to infiltrate our daily lives by exploiting the rapidity and reach of technology, as well as their expanding understanding of human behaviour and the brain. There is an increasing asymmetry in capability and agility, and that is why cognitive warfare is so dangerous.

For a moment, picture a future where influence operations are no longer limited to *what* you see but *when* you see it, and in what emotional state. Imagine I have access to your smartwatch data: your heart rate, stress levels, and sleep patterns. With this biometric insight, I do not need to guess when you are most vulnerable — I can know. And by gaining access to or manipulating the platforms that interpret this data, I can time the delivery of content to coincide with moments of heightened emotional or cognitive susceptibility.

If your heart rate spikes, signaling stress, fatigue, or agitation, I can inject targeted messaging into your environment: emotionally charged content, narrative reinforcements, or psychologically primed cues to shape your perception and behaviour. And I can do this at times when the data indicates your executive functioning is compromised: while mindlessly scrolling Instagram late at night, when your heart rate elevates after a fight with your spouse, or when your glucose levels are low before breakfast. This is not just targeting the *who* of influence; it is targeting the *when* and *how*. It enables real-time, individualized microtargeting that bypasses your rational defences and exploits your body's stress response as an entry point for manipulation.

The Engineering Behind Our Dysfunction

This future is not far off — it could well be tomorrow because, while our understanding of human behaviour and neuroscience is relatively young, it is maturing at a rapid pace.

One hundred years ago, we still believed in bloodletting. Fifty years ago, we institutionalized people for manageable conditions such as bipolar disorder. It wasn't until 2003 that the human genome was completely mapped. Over the past 20 to 30 years, our understanding of how the brain interacts with and shapes our realities has expanded rapidly in line with the parallel growth of technology. As a result, the ability to influence how societies function and individuals behave has also evolved.

With this understanding, malign and adversarial actors are not just observing our dysfunction; they are engineering it. And with the power of technology and the hours we dedicate to it, our identities are easily weaponized. The challenge is that many of us have been raised to believe that emotions and beliefs are something we push aside to get on with the business of the day. We grew up singing, "Sticks and stones may break my bones, but words will never hurt me."

Science tells us this is untrue. Words are enormously influential. Emotions, narratives, and worldviews can be powerful enough to make people protest their leaders, align with extremist ideologies, or engage in violent terrorism. What happens if entire populations reject long-held national values not because of genuine disagreement about the values themselves, but because adversaries have succeeded in changing prevailing views about how those values should be applied?

This is not hypothetical; this is the essence of cognitive warfare, and it is happening now. In this new world, society is the vector through which adversaries target political and military systems. They no longer need to target military forces directly if they can stir enough internal discord for citizens to turn against their governments, institutions, and alliances.

And waging this type of warfare is dangerously cost-effective — far cheaper than buying tanks or planes — precisely because it does not require physical force to achieve strategic effects. It targets not terrain but something far more fragile: truth, trust, and the will to act.

We Are All Vulnerable

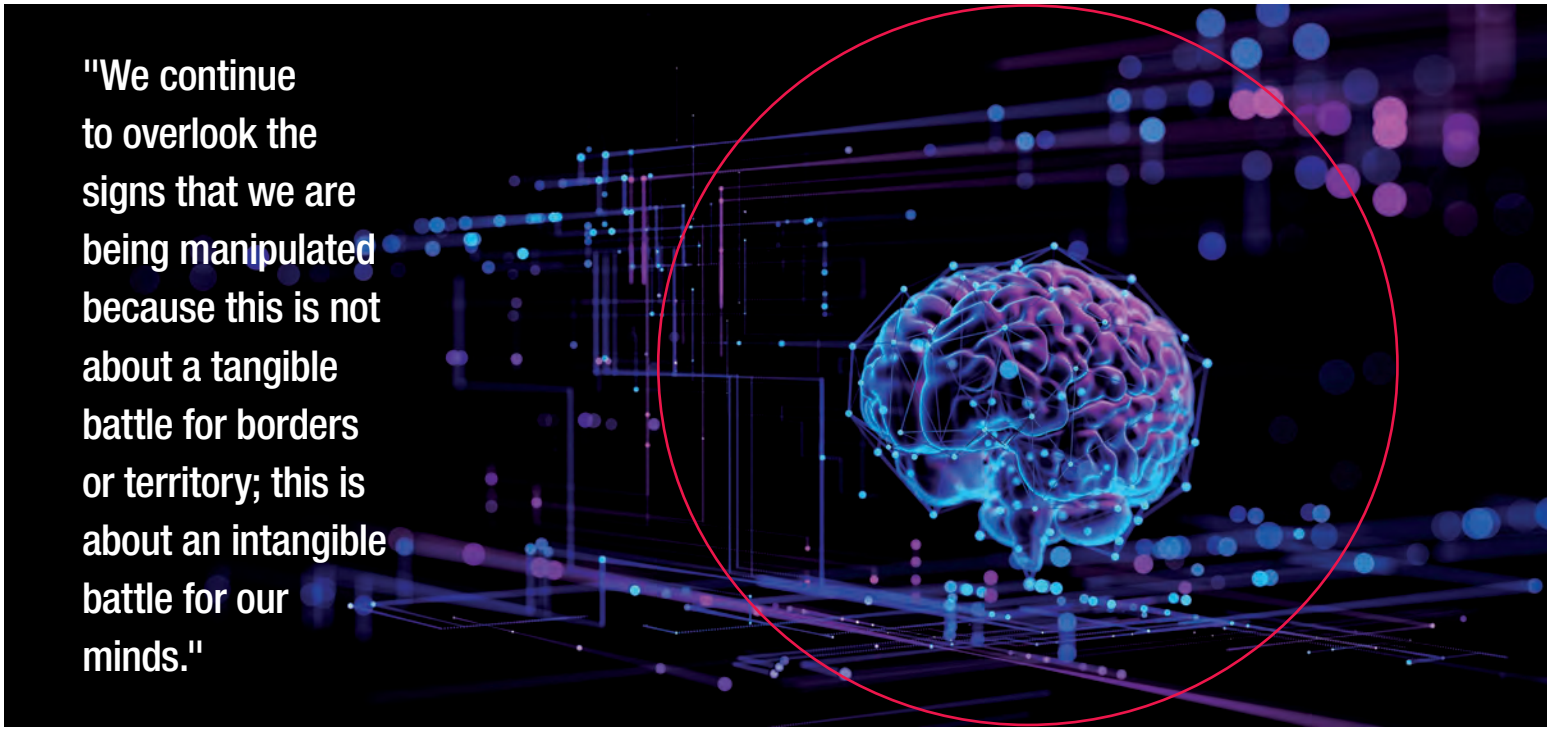
At a NATO Allied Command Transformation (ACT) cognitive warfare simulation event in 2023, national representatives were asked a simple question: Do you believe your country needs to address cognitive warfare? The majority said yes. However, when asked whether they felt personally vulnerable to cognitive warfare, very few did. Their responses were predictable, and they illustrate a fundamental point: how can we, as militaries, nations, and Allies, effectively counter and respond to cognitive warfare if we do not understand how we, as individuals, are vulnerable?

ROBUST PSYCHOLOGICAL and neuroscience research indicates that humans are predisposed to minimizing their own cognitive vulnerabilities while externalizing weaknesses onto others. This protective mechanism helps preserve self-confidence, but it also creates blind spots, impacting our ability to counter and respond to cognitive warfare. More often than not, we assume that the problem is other people: "I'm not vulnerable to influence; they are. I would recognize cognitive warfare."

This is not accurate.

Every individual is a target.

Twenty-four hours a day, our adversaries and competitors utilize trained specialists — and increasingly, machine cognition, AI, and other technologies — to analyze our media habits, affiliations, and identities, creating emotional and behavioural effects from afar. And their goal is clear: to prime and destabilize societies from within long before traditional confrontation occurs.



Real-World Examples

Between 2019 and 2021, ISIL-aligned insurgents in Mozambique's Cabo Delgado province conducted a systematic cognitive warfare campaign, weaponizing platforms such as Facebook, WhatsApp, and Telegram. They distributed graphic propaganda, fabricated footage, and false territorial claims to incite fear and evacuate towns before attacks occurred.[4] These efforts were paired with economic disruption, including strikes on energy infrastructure and supply chains, to amplify perceptions of state failure. Targeted assassinations of local leaders further eroded trust in governance, combining psychological intimidation with physical violence. The campaign glamourized fighters, denigrated the government, and attracted thousands of disenfranchised youths from across the region. By 2020, over 400,000 people were displaced, many fleeing due to fear fueled by online rumours rather than battlefield threats.[5] Cyber disruptions of humanitarian communications paralysed aid delivery, while selective kinetic strikes reinforced the illusion of militant omnipresence. These tactics decimated governance, derailed infrastructure projects, and disrupted humanitarian operations, all without large-scale combat.

China's growing use of AI in influence operations further demonstrates how adversaries are fusing emerging technologies with cognitive effects. Based on research from the New York Times, documents leaked from GoLaxy, a Chinese company tied to state security agencies, reveal how its "Smart Propaganda System" (GoPro) has been deployed in Hong Kong, Taiwan, and inside China to track debates, mine social media profiles, and generate targeted, adaptive propaganda that "feels authentic."[6]

In the 2024 Taiwanese elections, the system recommended narratives designed to exploit divisions in public opinion and weaken the Democratic Progressive Party. These efforts went beyond Russia-style troll farms, using AI to mass-produce and target content at scale. While not all operations proved decisive, the campaigns consistently sought to undermine trust in Taiwan's pro-independence leadership by reframing national identity and amplifying narratives of inevitability around Beijing's influence. GoLaxy's methods highlight the shift from time-intensive, handcrafted propaganda to AI-enabled identity manipulation, something branded as increasingly quicker, cheaper, and more targeted.

Currently, the company claims the ability to track over 180,000 X accounts in Hong Kong, monitor thousands of Western social media posts daily, and build virtual profiles on more than 2,000 U.S. political figures. In Taiwan, these capabilities mean that adversarial messaging can be explicitly aimed at citizens who are already primed by grievances over sovereignty and security. Even when content is factual, its emotional framing and algorithmic amplification create distorted perceptions of consensus and inevitability, crafting an environment where the ultimate target is not facts, but identity itself.[7]

What This Means for NATO

While Allies possess defensive tools such as regulation, legislation, and enforcement to respond to cognitive effects, these measures tend to be fragmented, reactive, and rather slow. This asymmetry in deterrence and defence allows adversaries to undermine societies, weakening national resilience and splintering NATO's collective defence posture.

The traditional visualization of NATO's military capability, holding strong on an eastern or southern flank to prevent a physical advance, is outdated and trapped in a 20th-century perspective.

Information has no borders. These new front lines will not appear on a map. The most straightforward and efficient path to defeating NATO's military capability is not by applying strength against strength but by degrading societal and political support for NATO's military actions before they begin. Our key vulnerabilities are no longer defined by the range and lethality of our weapons but by the openness of our societies.

A Whole-of-Society Problem

Unfortunately, the term "warfare" is somewhat misleading. Responding to cognitive warfare requires us to acknowledge that it cannot be addressed solely through military strength because it is primarily waged through the soft underbelly of society. Our least protected have become our easiest targets and most significant vulnerabilities.

The figure on page 35 illustrates this dynamic by depicting the symbiotic relationship between the military and civil society. The graphic demonstrates the dual nature of the cognitive warfare threat landscape, divided between the comparatively hardened military sphere and the vulnerable civil society domain.

"Information has no borders. Our key vulnerabilities are no longer defined by the range and lethality of our weapons but by the openness of our societies."

"NATO must do more than adapt — it must lead. This demands a full-spectrum, multi-domain approach focused on embedding cognitive warfare into doctrine, training, and operational planning."

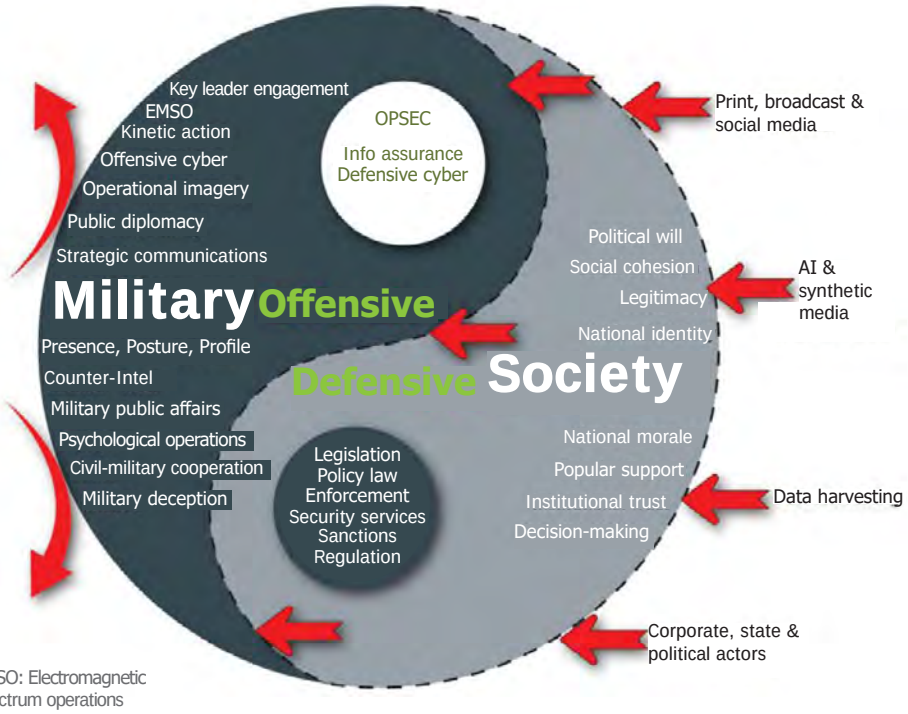
The red arrows represent threats in the information environment (IE) directed at the Alliance, aimed at achieving political and military objectives. The military half of the diagram illustrates traditional capabilities that protect the military instrument of power through tools aimed at adversaries, supported by layered defences such as information assurance, operational security, and defensive cyber. The military operates as a "hard target," fortified by doctrine, infrastructure, and threat awareness. While adversaries may attempt to disrupt military cognition and decision-making, such efforts can be anticipated and neutralized.

In contrast, the civil society half of the diagram is open, decentralized, and exposed. Democratic freedoms and a largely unregulated IE make it a "soft target," vulnerable to adversarial cognitive operations that manipulate perception, polarize opinion, and destabilize communities. These attacks exploit technologies such as AI, media ecosystems, and data harvesting. The dashed line around society reflects its porous boundaries, where adversaries target the accessible, unaware, and unprotected civil domain underpinning NATO's strength.

As a result, the psychological and societal effects of adversarial attacks can bleed into hardened military structures, especially in areas where the civil and military spheres intersect, eroding trust in defence institutions, weakening recruitment, and undermining public support. These attacks are not hypothetical. They degrade readiness, morale and legitimacy, threatening NATO's operational effectiveness.

Even if NATO fielded the strongest and best-resourced militaries in the world, it would not be enough if societies remained vulnerable. Adversaries target the foundations of resilience, knowing that no amount of military power can compensate for a fractured, unstable society. This spillover of effects between civil and military spheres has led to the misconception that cognitive warfare is incompatible with democratic values. However, like warfare in any domain, operations in the cognitive dimension reflect the strategic culture of the actors involved. Democracies need not abandon their principles; they must develop cognitive strategies aligned with their norms while countering adversaries who exploit openness and trust. Upholding democratic integrity while building resilience is not only possible but essential to safeguarding societal cohesion and long-term security.

Below
Graphic adapted from NATO Strategic Communications Centre of Excellence



EMSO: Electromagnetic spectrum operations

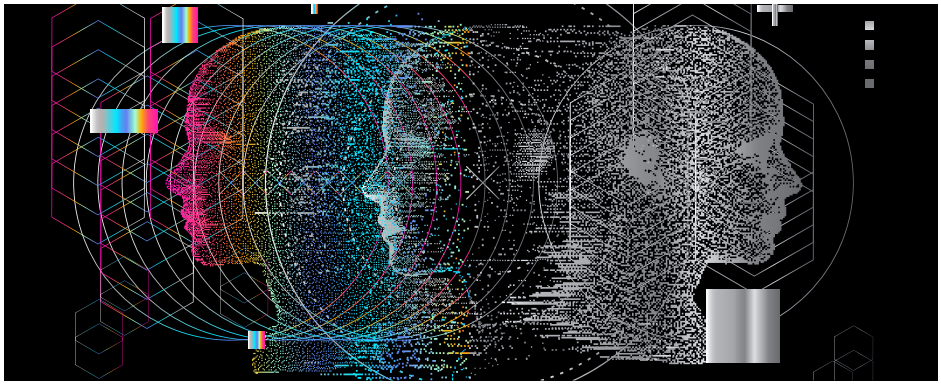
Maintaining Strategic Advantage: The NATO Cognitive Warfare Concept

States within the Alliance are making significant progress toward these goals, including implementing national and regional strategies to counter hostile information activities, enhancing media literacy, and supporting expanded research and development. However, we remain reactive, constrained by outdated legislative, systemic, and conceptual frameworks. The scale of current adversarial information proficiencies exceeds our capabilities — a strategic asymmetry that threatens our ability to act decisively in a crisis.

We face a stark choice: either invest in enhancing cognitive capabilities and adapting traditional perspectives on the role of the military and government within society, or accept that we will fall behind, risking failure to secure cognitive advantage in an increasingly hostile IE. Rapid and decisive action, including a societal paradigm shift, is required to defend our populations, institutions and military forces from degradation in the cognitive dimension.

To accomplish this, NATO must do more than adapt — it must lead. This demands a full-spectrum, multi-domain approach focused on embedding cognitive warfare into doctrine, training, and operational planning; resourcing influence capabilities and behavioural sciences; and uniting uniformed services with civilian institutions across the continuum of competition.

"Facts do not prevail on their own; emotion, identity, and resonance are just as powerful."



These ideas form the basis of NATO's Bilateral Strategic Command (Bi-SC) Cognitive Warfare Concept, a necessary and urgent call to action designed to address functional, legal, ethical, and doctrinal gaps and offer an actionable path forward. The concept acknowledges that cognitive warfare is no longer a supporting function — it is the contest itself. We must strengthen our ability to apply existing instruments of national power in a cohesive, integrated way across all phases of competition; something demonstrated during the early stages of Russia's 2022 invasion of Ukraine. While that unity has evolved with time, Ukraine continues to show what is possible when societies and militaries fight in concert, combining kinetic force with political, economic, diplomatic, and informational power. These hard-earned lessons must not be ignored.

From a strategic communications standpoint, we must also move beyond the assumption that "truth-telling" alone will win the battle for cognitive advantage.

If cognitive warfare has shown us anything, it is that facts do not prevail on their own; emotion, identity, and resonance are just as powerful, if not more so. It is not enough to broadcast our values and fact-check disinformation; we must also address the underlying issues that fuel it by engaging with the full spectrum of perspectives within our societies to deter adversarial weaponization of existing social discord.

Below
The author (right) was one of the speakers at the 2025 NATO Communicators Conference, which underscored that adversaries are no longer simply contesting the information space; they are deliberately targeting the Alliance's ability to think, decide, and act. Photo by HQ SACT PAO

Conclusion: We Are Already in the Fight

As individuals, military and otherwise, we must accept that we are already in the fight. But the front lines are not drawn on maps; they run through our institutions, our societies, and our minds. It is about all of us. There are no bombs, no borders — just the quiet hijacking of our perception. Our adversaries do not need to use expensive missiles and machinery if they can keep us distracted, divided, and emotionally reactive. They understand that influence does not require truth, but only our attention.

If we do not take steps to recognize the ubiquity and seriousness of the cognitive battlespace, which touches every aspect of our military and civilian lives, we will lose more than time; we will also lose readiness. We must invest tonight if we expect to fight tomorrow. If we fail to do so, we may wake up to discover that our thoughts are no longer our own. Not because we lost a war — but because we never appreciated that we were in one. ✧

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